NTNU – Trondheim
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TET4185: Power Markets, Resources and Environment
Organization of Electricity Markets

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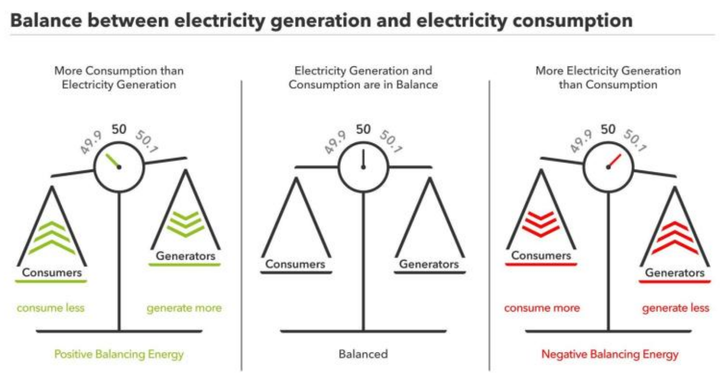
Overview

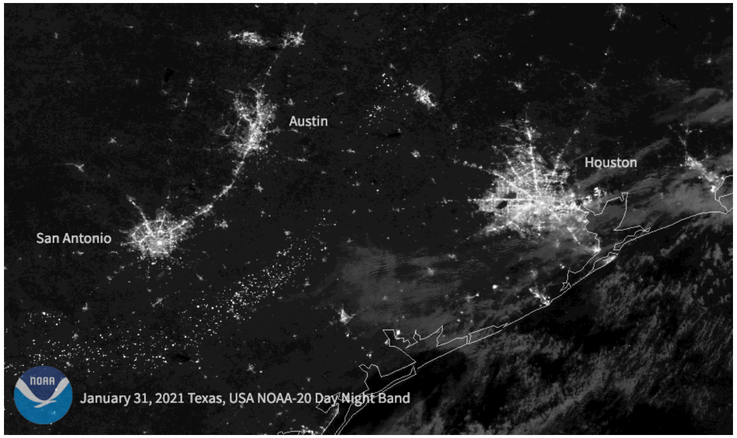
- Recap-Differences between electricity and other commodities
- Different types of electricity exchange
- The time-line of electricity market in the Nordic power market

Differences Between Electricity and Other Commodities

- Electricity is inextricably linked with a physical delivery system
  - Physical delivery system operates much faster than any market
  - Generation and load must be balanced at all times
  - Failure to balance leads to collapse of system
  - Economic consequences of collapse are enormous
  - Balance must be maintained at almost any cost
  - Physical balance cannot be left to a market



Source: https://www.next-kraftwerke.com/knowledge/afrr



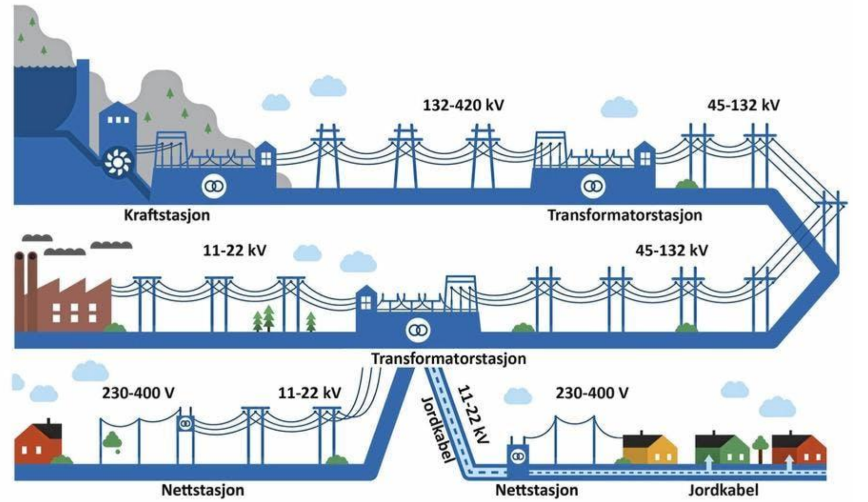
2021 Texas power crisis- At least 210 people lost their lives during the winter storm

Differences Between Electricity and Other Commodities

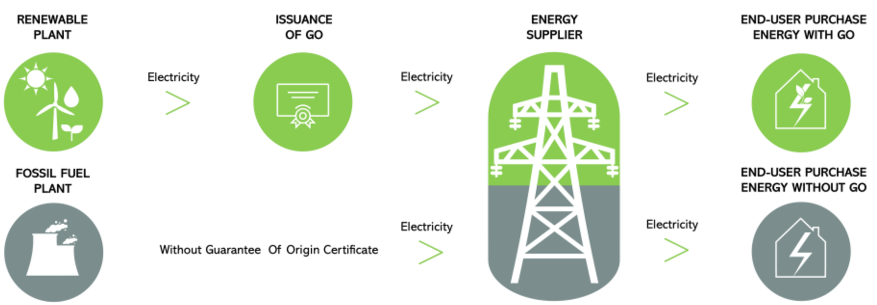
- Electricity generation is pooled
  - Electricity transferred through the grid is not physically labelled
  - Generators cannot choose their consumers and vice versa*
  - Electrical energy produced by all generators is indistinguishable

- Pooling is economically desirable

- A breakdown of the system affects everybody



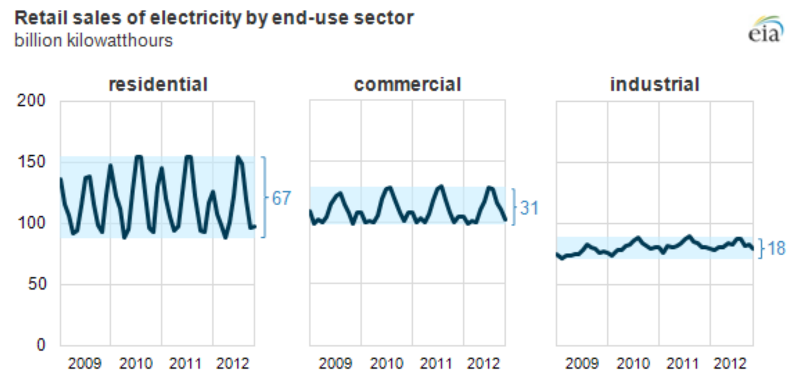
Source: https://www.bkk.no/om-bkk/bli-kjent-med-stromnettet



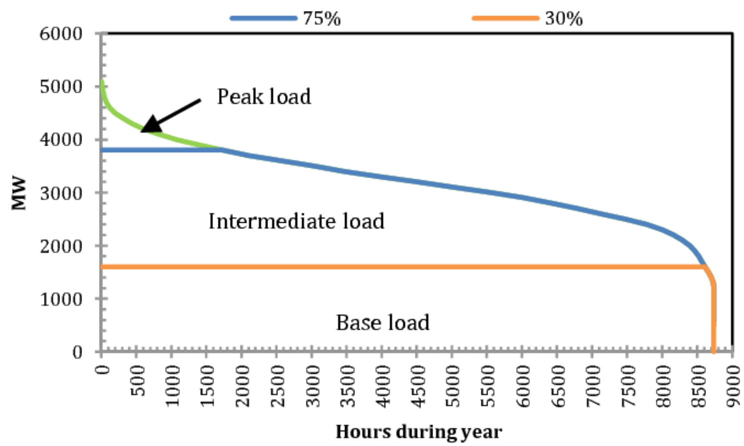
Source: https://www.aither.com/renewable-energy/guarantee-of-origin/

Differences Between Electricity and Other Commodities

- Electricity demand has predictable variation
  - Daily, weekly, seasonal variations
  - Similar to other commodities (e.g. coffee)
- Electricity cannot be stored
  - Must be consumed when produced
- Production must be able to deliver peak demand
  - The instantaneous power demand
- Short-term elasticity (change) of demand is very low
  - Demand curve is almost vertical



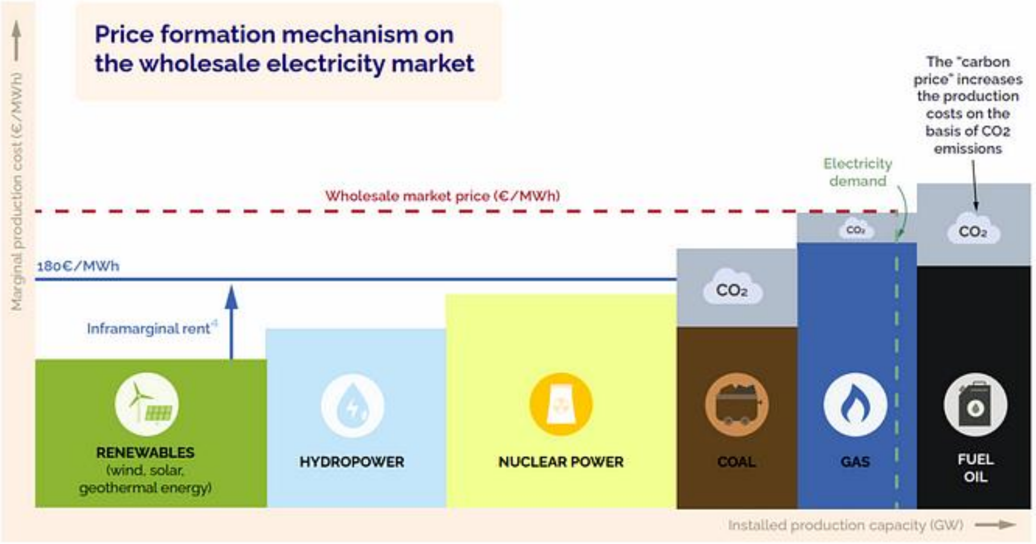
Source: https://www.eia.gov/todayinenergy/detail.php?id=10211



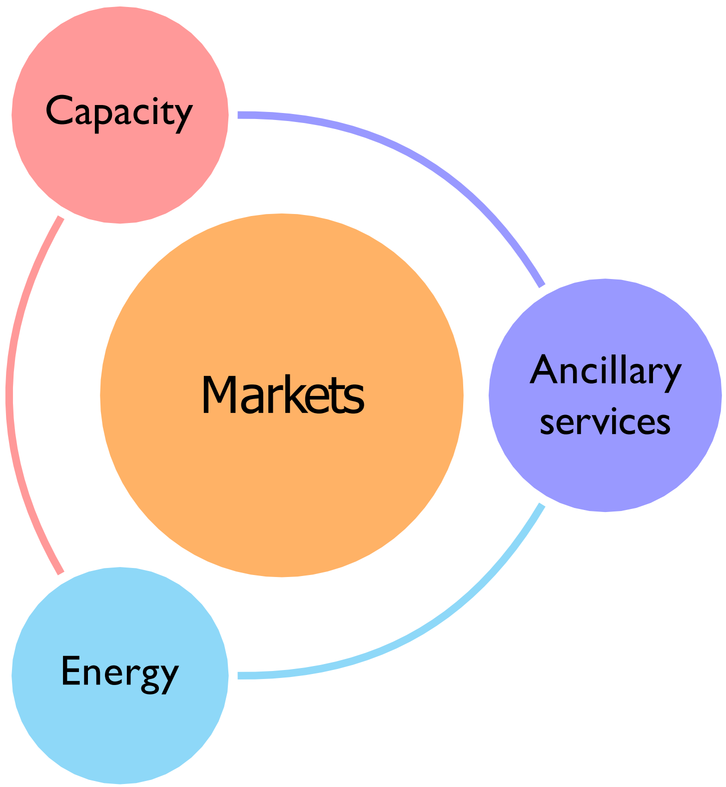
Source: Muslim, Hasan Noaman, Afaneen Alkhazraji, and Mohammed Ahmed Salih&. "Electrical Load Profile Analysis and Investigation of Baghdad City for." (2017).

Electricity Market

- A market is an environment to help buyers and sellers interact and agree on transactions
- Market equilibrium: the price clears the market, that is, the supply is equal to demand
- Large consumers and retailers purchase electrical energy from generating companies based on their forecasted consumption



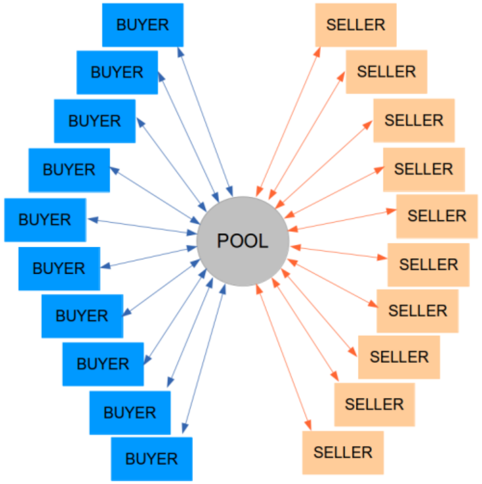
Different types of markets



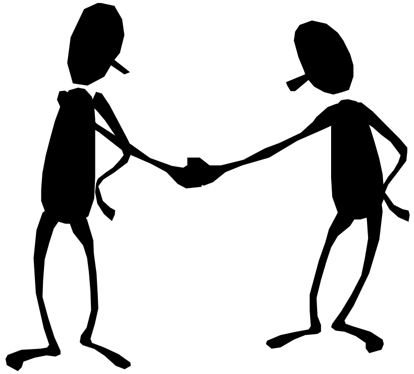
- **Capacity [MW]**: Sufficient power production in the grid. System operators trade this to ensure sufficient generation capacity is present for reliable system operation in future years and at competitive prices
- **Energy [MWh]**: a central place for the optimal scheduling and settlement of energy exchanges
- **Ancillary service [ΔMW]**: any type of service that supports power system operations, directly bought by the system operator, e.g.
  - Primary reserves (change within seconds)
  - Secondary reserves (change within minutes)
  - Tertiary reserves (slow change within minutes)
  - Black-start capability, short-circuit power, reactive reserves and voltage control

Different types of electricity exchange

- **Centralised trading (Auctions in an electricity pool)**
  - Every producer and consumer places offers and bids into a pool
  - No one knows about the others prior to clearing
  - Centralized market-clearing algorithm decides about bids and offers that are retained
  - Market is cleared, a price is found, specific offers/bids accepted

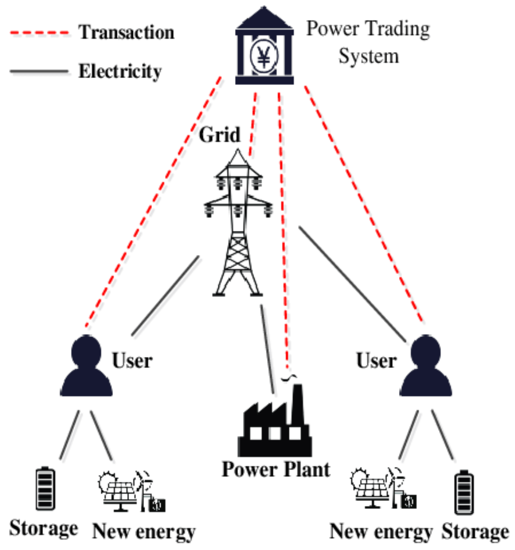


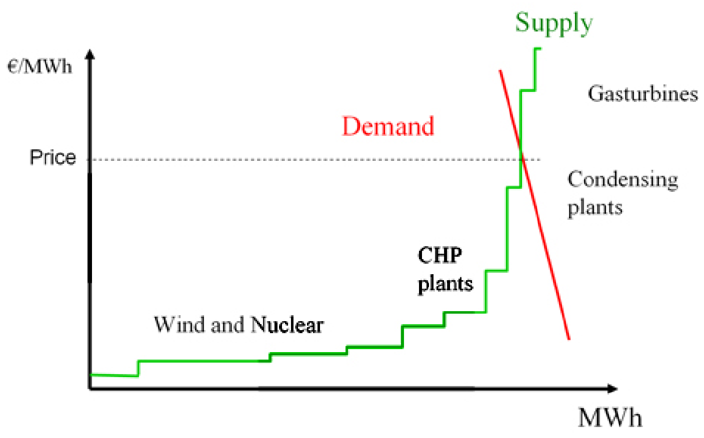
- **Bilateral trading (agreement between traders)**
  - This trading is for a direct exchange of power between a buyer and a seller
  - Does not use a pool-trading scheme, trades directly between each other
  - They may both be producers and /or consumers
  - Most likely, a broker is involved …



Centralised Trading

- All producers submit offers to the market operator
- All consumers submit bids to the market operator
- Market operator determines successful bids and offers and the Market Clearing Price
- In many electricity markets, the demand side is passive. A forecast of demand is used instead.





Example of centralised trading

Offers and bids in the Electricity Pool of Syldavia for the period from 9:00 till 10:00 on 11 June:

	Company	Quantity [MWh]	Price [$/MWh]
Offers	Red	200	12.00
	Red	50	15.00
	Red	50	20.00
	Green	150	16.00
	Green	50	17.00
	Blue	100	13.00
	Blue	50	18.00
Bids	Brown	50	13.00
	Brown	100	23.00
	Purple	50	11.00
	Purple	150	22.00
	Orange	50	10.0
	Orange	200	25.00

Example of centralised trading

Example of centralised trading

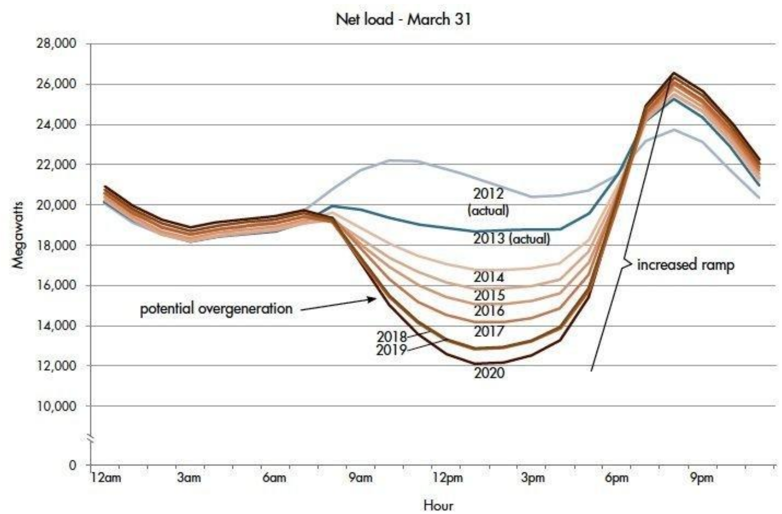
- Market clearing price: 16.00 $/MWh
- Volume traded: 450 MWh

Company	Production [MWh]	Consumption [MWh]	Revenue [$]	Expense [$]
Red	250		4000	
Blue	100		1600	
Green	100		1600	
Orange		200		3200
Brown		100		1600
Purple		150		2400
Total	**450**	**450**	**7200**	**7200**

Reasons for not treating each market period separately

- Why do we only consider 1 market time unit (MTU) of operation per market round?
- Should we treat these markets separately when bidding?

- Things to consider:
  - Operating constraints on generating units
    - How long can we operate/keep offline continuously? (down/up times)
    - How quickly can we change production? (ramping rates)
  - Saving achieved through scheduling
    - Cost of start-up and no-load operation
  - Reduce risk for generators
    - Uncertainty on the generation schedule leads to higher prices→ unit commitment



Source: https://www.energy.gov/eere/articles/confronting-duck-curve-how-address-over-generation-solar-energy

Power Pool or Power Exchange

Power Pool – US

- A central scheduling and dispatch
- Unit Commitment and Economic Dispatch
- Objective: Optimisation resources

Power Exchange- Nordpool

- It does not include an explicit optimisation
- The market participants are self-dispatch: determine themselves how to generate or consume depending on volumes and price
- Objective: Perfectly competitive market

Approaches to representing network constraints

- There are basically two philosophies, developed on both sides of the Atlantic:
  - US
  - Europe

	Europe	US
System Operator	TSO	ISO
Market Operator	Ind. Market Operator	ISO
Clearing	Supply-demand equilibrium	UCED problem
Prices	Zonal	Nodal

TSO: Transmission System Operator
ISO: Independent System Operator
UCED: Unit Commitment and Economic Dispatch

Bilateral trading-forward market

- Bilateral trading is the classical form of trading
- Involves only two parties:
  - Seller
  - Buyer
- Trading is a private arrangement between these parties
- Occasionally facilitated by brokers or electronic market operator
- Price and quantity negotiated directly between these parties- no “official” price

Pay-As-Bid

Types of bilateral trading depending on time scale

- Customized long-term contracts
- “Over the Counter” OTC trading
- Electronic trading

Customized long-term contracts

- Flexible terms
- Negotiated between the parties
- Duration of several months to several years

- Advantage:
  - Guarantees a fixed price over a long period
- Disadvantages:
  - Cost of negotiations is high

- Worthwhile only for large amounts of energy

OTC-“Over the Counter” trading

- Smaller amounts of energy
- Delivery according to standardized profiles
- Advantage:
  - Much lower transaction cost

- Used to refine position as delivery time approaches

Electronic trading (already leaning towards the pool concept)

- Buyers and sellers enter bids directly into computerized marketplace
- All participants can observe the prices and quantities offered, but participants remain anonymous
- Automatic matching of bids and offers
- Market operator handles the settlement

- Advantages:
  - Very fast and cheap
  - Very good source of information about the market

Example of bilateral trading

Generating units owned by Borduria Power:

Unit	P^{min} [MW]	P^{max} [MW]	MC [$/MWh]
A	100	500	10.0
B	10	200	13.0
C	0	50	17.0

Example of bilateral trading

Trades of Borduria Power for 11 June from 2:00 pm till 3:00 pm

Type	Contract Date	Identifier	Buyer	Seller	Amount [MWh]	Price [$/MWh]
Long term	10 January	LT1	Cheapo Energy	Borduria Power	200	12.5
Long term	7 February	LT2	Borduria Steel	Borduria Power	250	12.8
Future	3 March	FT1	Quality Electrons	Borduria Power	100	14.0
Future	7 April	FT2	Borduria Power	Perfect Power	30	13.5
Future	10 May	FT3	Cheapo Energy	Borduria Power	50	13.8

Net position: Sold 570 MW
Production capacity: 750 MW

Unit	P^{min} [MW]	P^{max} [MW]	MC [$/MWh]
A	100	500	10.0
B	10	200	13.0
C	0	50	17.0

Example of bilateral trading

(Electronic Trade)

Pending offers and bids on Borduria Power Exchange at mid-morning on 11 June for the period from 2:00 till 3:00 pm:

11 June 14:00-15:00	Identifier	Amount [MW]	Price [$/MWh]
Offers to sell energy	B5	20	17.50
	B4	25	16.30
	B3	20	14.40
	B2	10	13.90
	B1	25	13.70
Bids to buy energy	O1	20	13.50
	O2	30	13.30
	O3	10	13.25
	O4	30	12.80
	O5	50	12.55

Example of bilateral trading

Electronic trades made by Borduria Power:

11 June 14:00-15:00	Identifier	Amount [MW]	Price [$/MWh]
Offers to sell energy	B5	20	17.50
	B4	25	16.30
	B3	20	14.40
	B2	10	13.90
	B1	25	13.70
Bids to buy energy	O1	20	13.50
	O2	30	13.30
	O3	10	13.25
	O4	30	12.80
	O5	50	12.55

Net position: Sold 630 MW
Self schedule: Unit A: 500 MW
Unit B: 130 MW
Unit C: 0 MW

Example of bilateral trading

Unexpected problem: unit B can only generate 80 MW

Options:
- Do nothing and pay the balancing price for the missing energy
- Make up the deficit with unit C
- Trade on the available power exchange

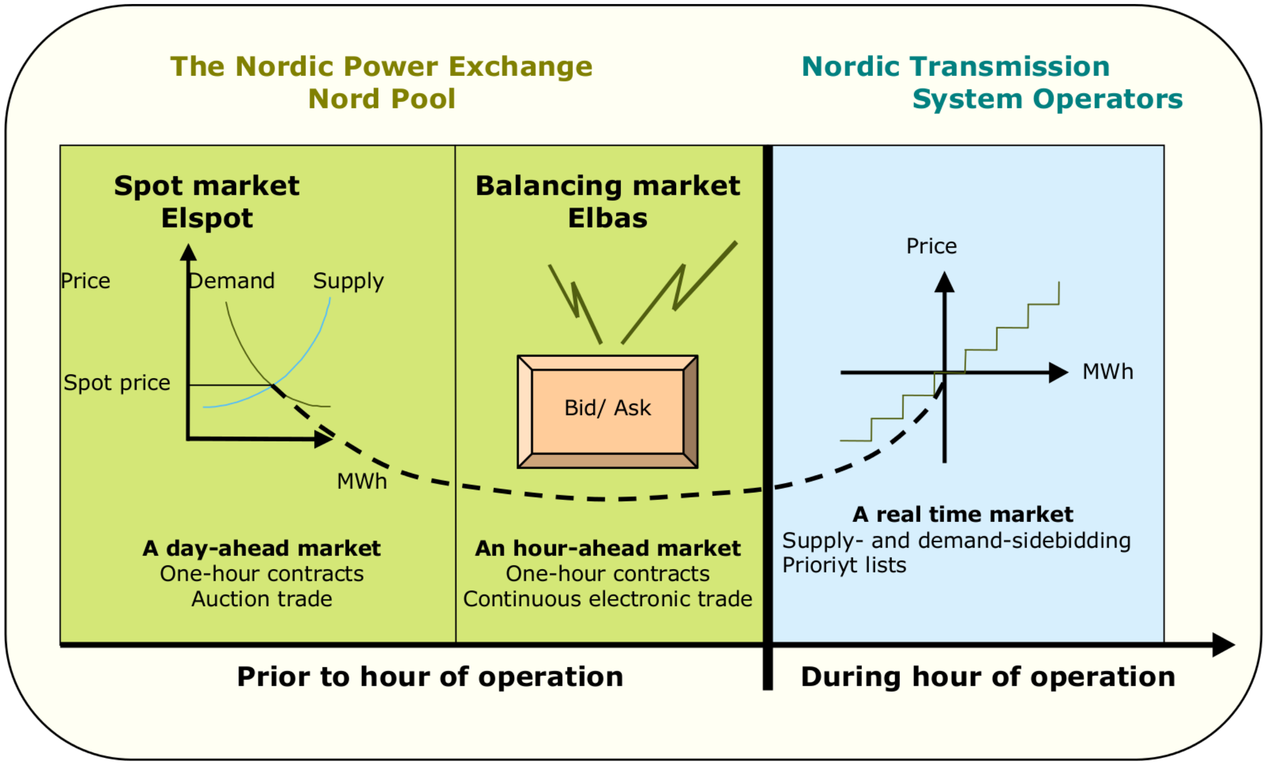
11 June 14:00-15:00	Identifier	Amount [MW]	Price [$/MWh]
Offers to sell energy	B5	20	17.50
	B4	25	16.30
	B3	20	14.40
	B6	20	14.30
	B8	10	14.10
Bids to buy energy	O4	30	12.80
	O6	25	12.70
	O5	50	12.55

Buying is cheaper than producing with C

New net position: Sold 580 MW

New schedule: A: 500 MW, B: 80 MW, C: 0 MW

The time-line of electricity market in the Nordic power market



Day-ahead (Elspot)

- A day-ahead market is a financial market!
  - These are only transactions - No one is “forced" to generate or consume...
  - Both market participants and system operator are informed about market clearing
  - outcomes (price and volumes for each market time unit)
  - In the European set-up, the market participants will then self-dispatch, i.e., determine themselves how they will generate or consume depending on volumes and prices

- The day-ahead market is cleared a reasonably long time before actual operations (between 12 and 36 hours)
- Remember there is a network involved, and power has to flow...

From day-ahead market to actual operation

- Alternative ways for the scheduled supply and demand to minimize imbalances:
  - Compensate with other generation/consumption means within their own portfolio

    → Re-dispatch of own units

  - Find ways to adjust through agreements with other players between the day-ahead market clearing and actual operations

    → Intra-day (/adjustment) market

  - Let the system operator put the system back to balance

    → Balancing market

Intra-day market (European Cross-Border Intraday (XBID))

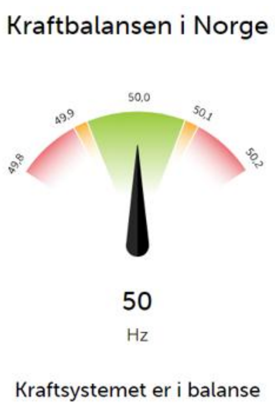
- While the day-ahead market is
  - a centralised trading
  - based on an auction mechanism
- The intraday market is based on bilateral contracts, even though centrally organized.
- Trading closes 60 minutes before delivery in most Nordic areas
- Some reasons for that:
  - less players,
  - less liquidity,
  - the need for corrective actions may highly vary depending upon how new information disclosure occurs between day-ahead market clearing and actual operations (XBID uses a single shared order book, a continuous trading system, and a common IT platform)...
  - Unlike day-ahead auctions, orders are matched continuously
- In Norway, XBID is operated through Nord Pool, with the Norwegian TSO Statnett providing grid capacity to the system.

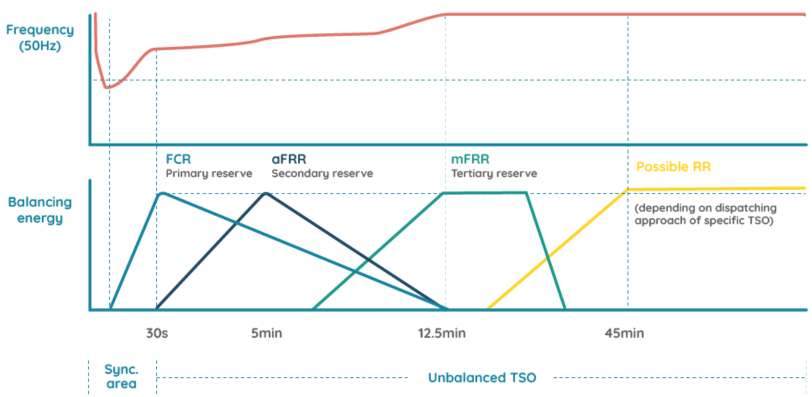
Balancing Market (frequency)

- Balancing market-single buyer
  - Provides a quick way of bridging the gap between supply and demand
  - Balancing market is used for adjustments of imbalances from day-ahead market clearing
  - **Balancing market is the market of last resort – A real-time market**

- System operators (TSO) manage this for each area (country-level, region etc.)
  - Maintain security-of-supply

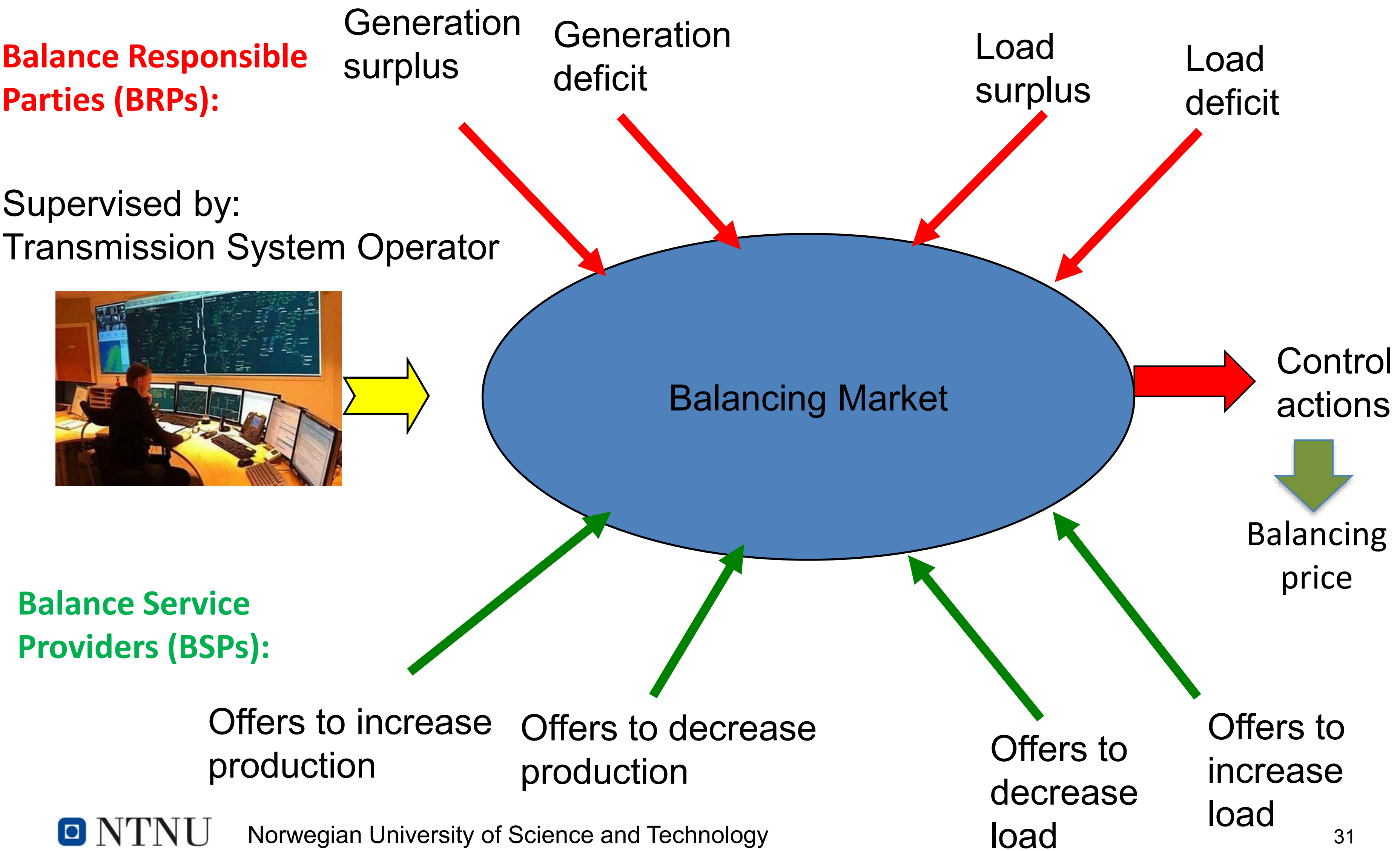
- Efficient balancing markets ensure the security of supply at a cost-reflective balancing prices





Source: https://sympower.net/the-differences-in-demand-response-balancing-services-explained/

Balancing market



Example of Balancing market

Borduria Power's position

Unit	P^{sched}	P^{min}	P^{max}	MC
	[MW]	[MW]	[MW]	[$/MWh]
A	500	100	500	10.0
B	80	10	80	13.0
C	0	0	50	17.0

- 11 June between 2:00 pm and 3:00 pm
- Balancing price: 18.25 $/MWh and Borduria is participating in balancing market using unit C.
- Unit B of Borduria Power could produce only 10 MWh instead of 80 MWh
- Borduria Power thus had a deficit of 70 MWh for this hour
- 40 MW of Borduria Power's balancing market bid was called by the system operator

Settlement Process

The process of getting money from people who bought energy to the people who have produced it

- Day-ahead market (Centralised trading)
  - Market operator collects from consumer
  - Market operator pays producers
  - All energy traded at the Market Clearing Price (MCP)

- Intraday Market (Bilateral trading)
  - Bilateral trades settled directly by the parties as if they had been performed exactly

- Balancing Market (All imbalances in a given settlement period are priced at the same balancing price →single pricing)
  - If system is in short position
    - Produced more or consumed less → received balancing price
    - Produced less or consumed more→ pay balancing price

Borduria Power's Settlement (ex post)

Market	Type	Amount [MWh]	Price [$/MWh]	Revenues [$]	Expenses [$]
Futures and Forwards	Sale	200	12.50	2,500.00	
	Sale	250	12.80	3,200.00	
	Sale	100	14.00	1,400.00	
	Purchase	-30	13.50		405.00
	Sale	50	13.80	690.00	
Power Exchange	Sale	20	13.50	270.00	
	Sale	30	13.30	399.00	
	Sale	10	13.25	132.50	
	Purchase	-20	14.40		288.00
	Purchase	-20	14.30		286.00
	Purchase	-10	14.10		141.00
Balancing Market	Sale	40	18.25	730.00	
	Imbalance	-70	18.25		1,277.50
Production Cost	Unit A	-500	10.00		5,000.00
	Unit B	-10	13.00		130.00
	Unit C	-40	17.00		680.00
Total		0		**9,321.50**	**8,207.50**